

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

CourseCode: 23CM4121

ExperimentNo: 1&2

StudentName: Tammina Ushasri

RollNo:A24126552120

Branch:CSM

Date: 08-09-2026

LABORATORY OBSERVATION / RECORD

1. TITLE

Implement Modules in NodeJS

2. AIM

To implement and use modules in NodeJS by creating a separate module file, exporting functions using `module.exports`, and importing them using `require()`.

3. OBJECTIVES

- To understand the concept of modules in NodeJS.
- To create a user-defined module.
- To export functions using `module.exports`.
- To import a module using `require()`.
- To perform addition and subtraction using exported functions.
- To execute and verify the NodeJS program.

4. THEORY

A **module** in NodeJS is a reusable piece of code that can be used in different programs. Modules help organize code into separate files and make programs easier to maintain.

In this program, `math.js` is created as a user-defined module. It contains two functions: `add()` and `subtract()`.

The `module.exports` object is used to export these functions from `math.js`. The `require()` function is then used in `app.js` to import the `math.js` module.

Important concepts:

- `module.exports` – exports functions or values from a module.
- `require()` – imports a module into another JavaScript file.

- **User-defined module** – a module created by the programmer.

5. ALGORITHM / PROCEDURE

Program: Implement Modules in NodeJS

1. Start.
2. Create a file named math.js.
3. Define the add() function.
4. Define the subtract() function.
5. Export both functions using module.exports.
6. Create another file named app.js.
7. Import math.js using require().
8. Assign values 20 and 10 to variables a and b.
9. Call the add() function.
10. Call the subtract() function.
11. Display the results using console.log().
12. Run the program using NodeJS.
13. Stop.

6. PROGRAM

File 1: math.js

```
JS math.js > ...
1  // math.js
2
3  function add(a, b) {
4    |   return a + b;
5  }
6
7  function subtract(a, b) {
8    |   return a - b;
9  }
10
11  module.exports = {
12    |   add,
13    |   subtract
14  };
```

File 2:app.js

```
JS app.js > ...
1 // app.js
2
3 const math = require("./math");
4
5 let a = 20;
6 let b = 10;
7
8 console.log("Addition =", math.add(a, b));
9 console.log("Subtraction =", math.subtract(a, b));|
```

Folder Structure

```
node-modules/
├──
├── math.js
└── app.js
```

Execution Command

node app.js

7. CODE EXPLANATION

Code	Explanation
function add(a, b)	Defines a function to add two numbers.
return a + b;	Returns the addition result.
function subtract(a, b)	Defines a function to subtract two numbers.
return a - b;	Returns the subtraction result.
module.exports = { add, subtract };	Exports both functions from math.js.
require("./math")	Imports the math.js module into app.js.
let a = 20;	Stores the first number.
let b = 10;	Stores the second number.
math.add(a, b)	Calls the imported addition function.
math.subtract(a, b)	Calls the imported subtraction function.
console.log()	Displays the result in the terminal.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	a = 20, b = 10	Addition should be 30	Addition = 30	Pass
TC-02	a = 20, b = 10	Subtraction should be 10	Subtraction = 10	Pass
TC-03	Import math.js using require()	Module should be imported successfully	Module imported	Pass
TC-04	Execute node app.js	Program should run without errors	Program executed	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
			successfully	

Execution Data

Input:

a = 20
b = 10

Command:

node app.js

Expected Output:

Addition = 30
Subtraction = 10

9. OUTPUT

Addition = 30
Subtraction = 10

10. RESULT

Result: Thus, modules in NodeJS were successfully implemented using module.exports and require(). The add() and subtract() functions were exported from math.js, imported into app.js, and executed successfully with the expected output.

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\AISHWARYA\OneDrive\Desktop\clg information portal> node app.js
Addition = 30
Subtraction = 10
PS C:\Users\AISHWARYA\OneDrive\Desktop\clg information portal> 
```